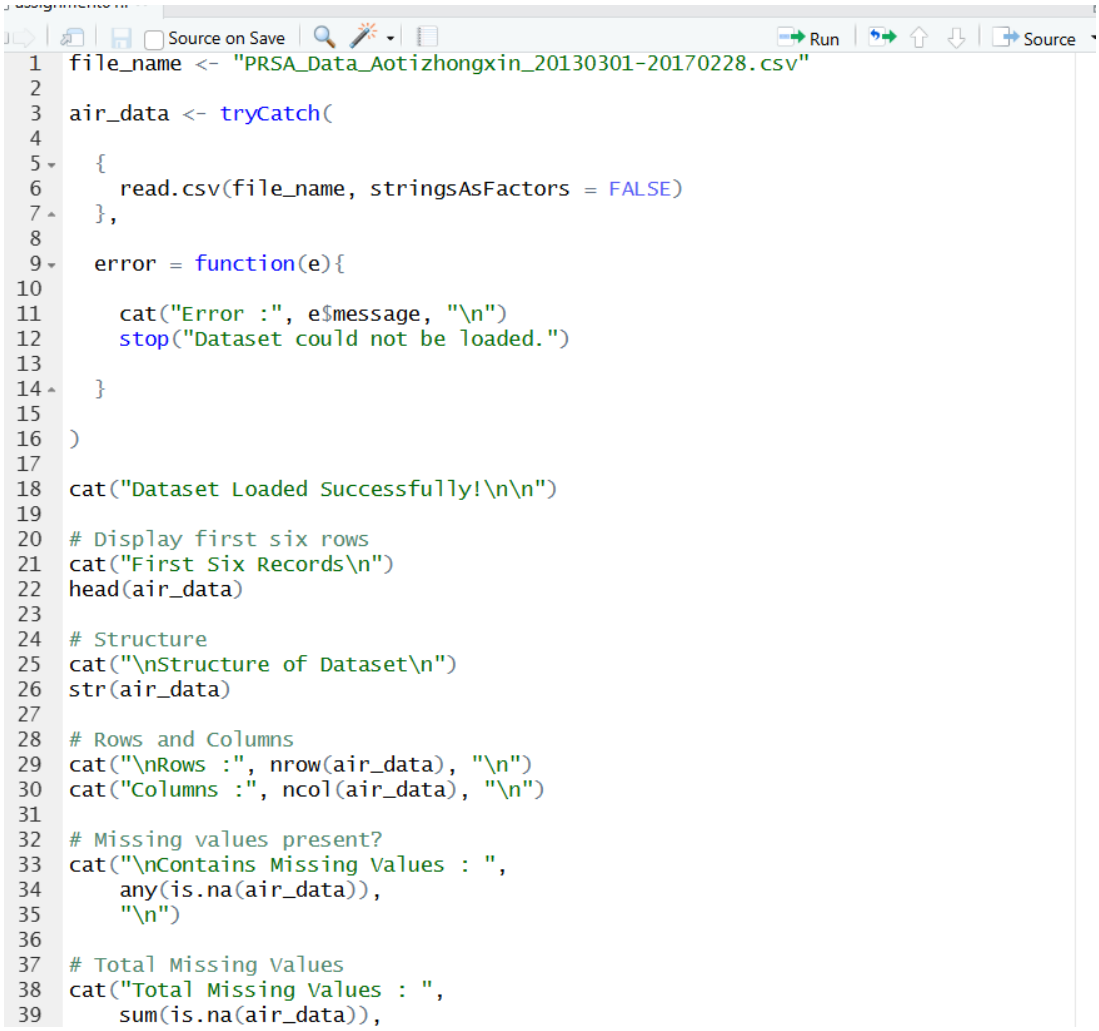
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Sub : R Programming

Experiment – 01



```r
file_name <- "PRSA_Data_Aotizhongxin_20130301-20170228.csv"

air_data <- tryCatch(

  {
    read.csv(file_name, stringsAsFactors = FALSE)
  },

  error = function(e){

    cat("Error :", e$message, "\n")
    stop("Dataset could not be loaded.")

  }

)

cat("Dataset Loaded Successfully!\n\n")

# Display first six rows
cat("First Six Records\n")
head(air_data)

# Structure
cat("\nStructure of Dataset\n")
str(air_data)

# Rows and Columns
cat("\nRows :", nrow(air_data), "\n")
cat("Columns :", ncol(air_data), "\n")

# Missing values present?
cat("\nContains Missing Values : ",
    any(is.na(air_data)),
    "\n")

# Total Missing Values
cat("Total Missing Values : ",
    sum(is.na(air_data)),
```

```r
    "\n")

# Task 2 : NA, NULL and NaN

# NA Example
temperature <- c(28, 30, NA, 32)

# NULL Example
missing_object <- NULL

# NaN Example
undefined_value <- 0 / 0

cat("\nTemperature Vector:\n")
print(temperature)

cat("\nCheck NA:\n")
print(is.na(temperature))

cat("\nNULL Object:\n")
print(missing_object)

cat("is.null(): ")
print(is.null(missing_object))

cat("\nNaN Value:\n")
print(undefined_value)

cat("is.nan(): ")
print(is.nan(undefined_value))


# Task 3 : Missing Summary Function

missing_summary <- function(df)
{

  variables <- c("PM2.5","PM10","SO2","NO2","TEMP","WSPM","wd")

```

```r
  summary_table <- data.frame(
    Variable=character(),
    Total_Records=integer(),
    Missing_Values=integer(),
    Missing_Percentage=double(),
    stringsAsFactors=FALSE
  )

  for(var in variables)
  {

    total <- nrow(df)

    missing <- sum(is.na(df[[var]]))

    percent <- (missing/total)*100

    summary_table <- rbind(summary_table,
                           data.frame(
                             Variable=var,
                             Total_Records=total,
                             Missing_Values=missing,
                             Missing_Percentage=round(percent,2)
                           ))

    if(percent > 20)
    {
      warning(paste(var,
                    "contains more than 20% missing values"))
    }

  }

  return(summary_table)

}

summary_result <- missing_summary(air_data)

```

```r
cat("\nMissing Value Summary\n\n")

print(summary_result)

# Task 4 : Identify Invalid Numerical Results

# Create pollution_ratio
air_data$pollution_ratio <- air_data$PM2.5 / air_data$PM10

# Count invalid values
na_count <- sum(is.na(air_data$pollution_ratio))
nan_count <- sum(is.nan(air_data$pollution_ratio))
pos_inf <- sum(air_data$pollution_ratio == Inf, na.rm = TRUE)
neg_inf <- sum(air_data$pollution_ratio == -Inf, na.rm = TRUE)

cat("NA Values :", na_count, "\n")
cat("NaN Values :", nan_count, "\n")
cat("Positive Infinity :", pos_inf, "\n")
cat("Negative Infinity :", neg_inf, "\n")

# Replace NaN and Infinite values with NA
air_data$pollution_ratio[is.nan(air_data$pollution_ratio)] <- NA
air_data$pollution_ratio[is.infinite(air_data$pollution_ratio)] <- NA

cat("\nInvalid values replaced successfully.\n")

# Task 5 : Replace Missing Numerical Values

numeric_variables <- c(
  "PM2.5",
  "PM10",
  "SO2",
  "NO2",
  "TEMP",
  "WSPM"
)

# Store missing values before cleaning
missing_before <- c()
```

```r

for(var in numeric_variables)
{

  if(var %in% names(air_data))
  {

    before <- sum(is.na(air_data[[var]]))

    median_value <- median(
      air_data[[var]],
      na.rm = TRUE
    )

    air_data[[var]][is.na(air_data[[var]])] <- median_value

    after <- sum(is.na(air_data[[var]]))

    missing_before[var] <- before

    cat("\nVariable :", var, "\n")
    cat("Missing Before :", before, "\n")
    cat("Median Used :", median_value, "\n")
    cat("Missing After :", after, "\n")

  }
  else
  {
    cat(var,"does not exist.\n")
  }

}

# Task 6 : Replace Missing Categorical Values

calculate_mode <- function(x)
{

  x <- x[!is.na(x)]
```

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```r

  unique_values <- unique(x)

  unique_values[
    which.max(tabulate(match(x, unique_values)))
  ]

}

before_wd <- sum(is.na(air_data$wd))

mode_wd <- calculate_mode(air_data$wd)

air_data$wd[is.na(air_data$wd)] <- mode_wd

after_wd <- sum(is.na(air_data$wd))

cat("Mode of wd :", mode_wd, "\n")
cat("Missing Before :", before_wd, "\n")
cat("Missing After :", after_wd, "\n")

# Task 7 : Error Handling Function

clean_variable <- function(df, variable_name)
{
  tryCatch({

    # Check if variable exists
    if(!(variable_name %in% names(df)))
    {
      stop("Variable does not exist.")
    }

    # Check if numerical
    if(!is.numeric(df[[variable_name]]))
    {
      stop("Variable is not numerical.")
    }

```

```r
    # Check if all values are missing
    if(all(is.na(df[[variable_name]])))
    {
      stop("Variable contains only missing values.")
    }

    median_value <- median(df[[variable_name]], na.rm = TRUE)

    if(is.na(median_value))
    {
      stop("Median could not be calculated.")
    }

    df[[variable_name]][is.na(df[[variable_name]])] <- median_value

    cat(variable_name,"cleaned successfully.\n")

    return(df[[variable_name]])

  },

  error = function(e)
  {
    cat("Error :",e$message,"\n")
    return(NULL)
  })

}

# Test Cases

clean_variable(air_data,"PM2.5")

clean_variable(air_data,"wd")

clean_variable(air_data,"ABC")


# Task 8 : Comparison Table
```

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```r
variables <- c(
  "PM2.5",
  "PM10",
  "SO2",
  "NO2",
  "TEMP",
  "WSPM",
  "wd"
)

missing_after <- c()

for(v in variables)
{
  missing_after[v] <- sum(is.na(air_data[[v]]))
}

comparison_table <- data.frame(

  Variable = variables,

  Missing_Before = c(
    missing_before["PM2.5"],
    missing_before["PM10"],
    missing_before["SO2"],
    missing_before["NO2"],
    missing_before["TEMP"],
    missing_before["WSPM"],
    before_wd
  ),

  Missing_After = c(
    missing_after["PM2.5"],
    missing_after["PM10"],
    missing_after["SO2"],
    missing_after["NO2"],
    missing_after["TEMP"],
    missing_after["WSPM"],
    missing_after["wd"]
```

353:3 (Top Level)

```r
  )

)

comparison_table$Values_Replaced <-
  comparison_table$Missing_Before -
  comparison_table$Missing_After

print(comparison_table)



cat("\nInterpretation:\n")

if (all(comparison_table$Missing_After == 0)) {
  cat("All selected missing values have been handled successfully.\n")
} else {
  cat("Some missing values still remain.\n")
}

# Task 9 : Visualization

before_values <- comparison_table$Missing_Before

after_values <- comparison_table$Missing_After

plot_data <- rbind(before_values,after_values)

barplot(

  plot_data,

  beside=TRUE,

  col=c("red","darkgreen"),

  names.arg=comparison_table$Variable,

  main="Missing Values Before and After Cleaning",
```

```
353     xlab="Variables",
354
355     ylab="Number of Missing Values",
356
357     legend.text=c("Before","After")
358
359 )
360
361 # Task 10 : Export Dataset
362
363
364
365 write.csv(
366
367     air_data,
368
369     "cleaned_air_quality_data.csv",
370
371     row.names=FALSE
372
373 )
374
375 cat("Dataset exported successfully.\n")
376
```

353:3 (Top Level)

Output :

```
> source("C:/assignment_work/sem7/R/assigment1/assignment01.r")
Dataset Loaded Successfully!

First Six Records

Structure of Dataset
'data.frame':   35064 obs. of  18 variables:
 $ No     : int  1 2 3 4 5 6 7 8 9 10 ...
 $ year   : int  2013 2013 2013 2013 2013 2013 2013 2013 2013 2013 ...
 $ month  : int  3 3 3 3 3 3 3 3 3 3 ...
 $ day    : int  1 1 1 1 1 1 1 1 1 1 ...
 $ hour   : int  0 1 2 3 4 5 6 7 8 9 ...
 $ PM2.5  : num  4 8 7 6 3 5 3 3 3 3 ...
 $ PM10   : num  4 8 7 6 3 5 3 6 6 8 ...
 $ SO2    : num  4 4 5 11 12 18 18 19 16 12 ...
 $ NO2    : num  7 7 10 11 12 18 32 41 43 28 ...
 $ CO     : int  300 300 300 300 300 400 500 500 500 400 ...
 $ O3     : num  77 77 73 72 72 66 50 43 45 59 ...
 $ TEMP   : num  -0.7 -1.1 -1.1 -1.4 -2 -2.2 -2.6 -1.6 0.1 1.2 ...
 $ PRES   : num  1023 1023 1024 1024 1025 ...
 $ DEWP   : num  -18.8 -18.2 -18.2 -19.4 -19.5 -19.6 -19.1 -19.1 -19.2 -19.3 ...
 $ RAIN   : num  0 0 0 0 0 0 0 0 0 0 ...
 $ wd     : chr  "NNW" "N" "NNW" "NW" ...
 $ WSPM   : num  4.4 4.7 5.6 3.1 2 3.7 2.5 3.8 4.1 2.6 ...
 $ station: chr  "Aotizhongxin" "Aotizhongxin" "Aotizhongxin" "Aotizhongxin" ...

Rows : 35064
Columns : 18

Contains Missing Values :  TRUE
Total Missing Values :  7271

Temperature Vector:
[1] 28 30 NA 32
```

R 4.6.1 · C:/assignment_work/sem7/R/assigment1/

```
Check NA:
[1] FALSE FALSE  TRUE FALSE

NULL Object:
NULL
is.null(): [1] TRUE

NaN Value:
[1] NaN
is.nan(): [1] TRUE

Missing Value Summary

  Variable Total_Records Missing_Values Missing_Percentage
1    PM2.5         35064            925                2.64
2     PM10         35064            718                2.05
3      SO2         35064            935                2.67
4      NO2         35064           1023                2.92
5     TEMP         35064             20                0.06
6     WSPM         35064             14                0.04
7       wd         35064             81                0.23
NA Values : 934
NaN Values : 0
Positive Infinity : 0
Negative Infinity : 0

Invalid values replaced successfully.

Variable : PM2.5
Missing Before : 925
Median Used : 58
Missing After : 0

Variable : PM10
```

```
Variable : PM10
Missing Before : 718
Median Used : 87
Missing After : 0

Variable : SO2
Missing Before : 935
Median Used : 9
Missing After : 0

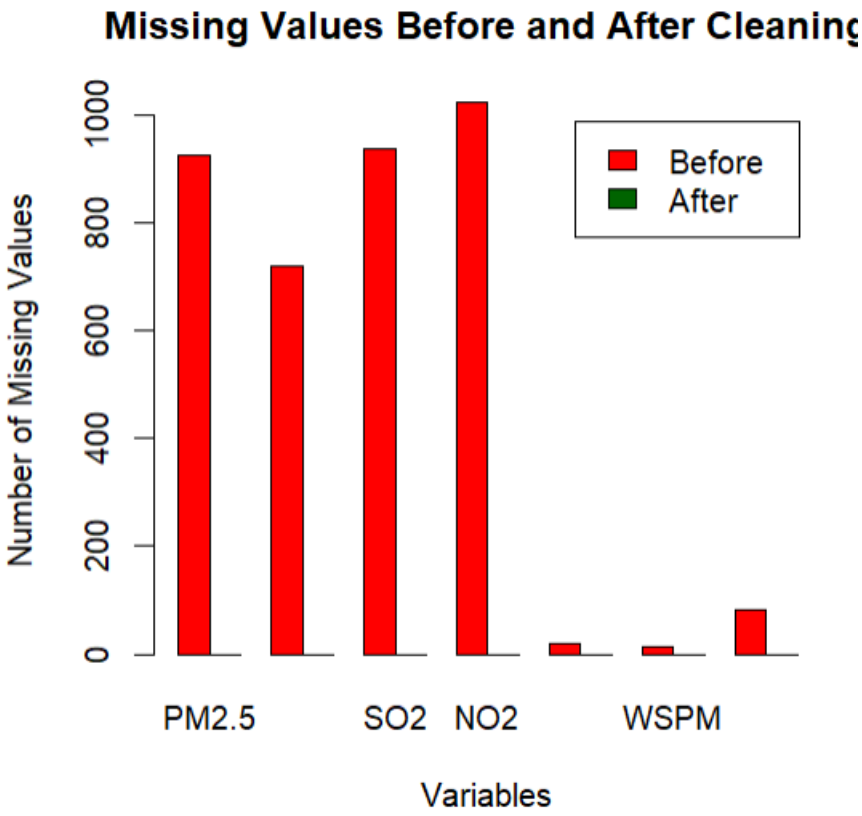
Variable : NO2
Missing Before : 1023
Median Used : 53
Missing After : 0

Variable : TEMP
Missing Before : 20
Median Used : 14.5
Missing After : 0

Variable : WSPM
Missing Before : 14
Median Used : 1.4
Missing After : 0
Mode of wd : NE
Missing Before : 81
Missing After : 0
PM2.5 cleaned successfully.
Error : Variable is not numerical.
Error : Variable does not exist.
```

	Variable	Missing_Before	Missing_After	Values_Replaced
PM2.5	PM2.5	925	0	925
PM10	PM10	718	0	718
SO2	SO2	935	0	935
NO2	NO2	1023	0	1023
TEMP	TEMP	20	0	20
WSPM	WSPM	14	0	14
	wd	81	0	81

```
Interpretation:
All selected missing values have been handled successfully.
Dataset exported successfully.
>
```


Missing Values Before and After Cleaning
Number of Missing Values
0
200
400
600
800
1000
Before
After
PM2.5
SO2
NO2
WSPM
Variables